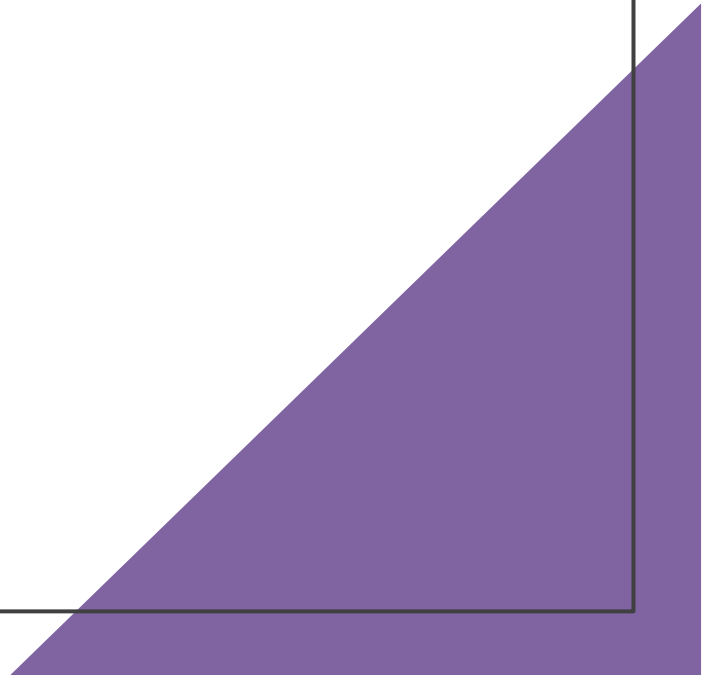


# NLP Processing Pipeline

- Raw Responses (1,120)
  - Text Cleaning (whitespace, line breaks, preserve wording)
  - Word Count
  - Rule-based Feature Extraction (Regex + Keywords)
    - Detect Phone Numbers, Crisis Short Codes & URLs
    - Named Entity Recognition (spaCy)
    - Cultural Coding (Professional, Family, Community, Religion)
    - Sentiment Analysis (VADER)
  - Specificity Score = Phone + Short Codes + URLs
- Structured Coding Matrix → Statistical Analysis



# NLP Feature Extraction

## Automatic NLP Feature Extraction

For each response  $r_i$ , a feature vector was generated:  
where

$$\mathbf{x}_i = (w_i, p_i, u_i, s_i, c_i, f_i, r_i, m_i, t_i)$$

- $w_i$  = word count
- $p_i$  = phone numbers detected
- $u_i$  = URLs detected
- $s_i$  = emergency short codes
- $c_i$  = crisis-number flag
- $f_i$  = family references
- $r_i$  = religion/spirituality references
- $m_i$  = professional support references
- $t_i$  = community support references

# Word Count & Regex Extraction

- - Word Count =  $|\text{tokens}(\text{response})|$  , Implementation : `len(clean_text.split())`
- - Phone Count =  $\sum I(\text{phone regex})$   $I(x) = \begin{cases} 1, & \text{if detected} \\ 0, & \text{otherwise} \end{cases}$
- - URL Count =  $\sum I(\text{URL regex})$
- - Emergency Short Code Detection
- - Short CodeCount =  $\sum I(\text{emergency code})$
- Dictionary :
  - 999
  - 911
  - 988
  - 112
  - 111
  - ...

# Dictionary-based Theme Extraction

- $\text{Theme}_k = \sum I(\text{keyword}_j)$
- Religion: God, prayer, church, mosque, faith...
- Family: family, mother, father, sibling...
- Professional: therapist, psychologist, GP...
- Community: friend, support group, volunteer...

| Step | Method (Abbreviation)                                            | Formula / Output                                                  | used to                                                                                                    |
|------|------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 1    | Text Cleaning                                                    | Standardised text                                                 | Remove formatting inconsistencies while preserving the original wording.                                   |
| 2    | Word Count ( <b>WC</b> )                                         | $(WC = \sum_{i=1}^n 1)$                                           | Measure response length and level of detail.                                                               |
| 3    | Regular Expression Matching ( <b>Regex</b> )                     | Pattern matching                                                  | Detect telephone numbers, crisis short codes and URLs automatically.                                       |
| 4    | Rule-Based Keyword Matching ( <b>RBKM</b> )                      | Dictionary matching                                               | Identify professional, family, community and religious support recommendations.                            |
| 5    | Named Entity Recognition ( <b>NER</b> )                          | spaCy entity extraction                                           | Identify organisations (ORG), locations (GPE), people (PERSON), etc., to assess localisation of responses. |
| 6    | Valence Aware Dictionary and sEntiment Reasoner ( <b>VADER</b> ) | Compound score (-1 to +1)                                         | Measure the emotional tone of each response using a lexicon- and rule-based approach.                      |
| 7    | Specificity Score ( <b>SS</b> )                                  | $(SS = N_{\text{phone}} + N_{\text{shortcode}} + N_{\text{URL}})$ | Measure how actionable each response is by counting concrete support resources.                            |
| 8    | Binary Feature Coding ( <b>BFC</b> )                             | (1=) Present, (0=) Absent                                         | Convert qualitative features into quantitative variables for statistical analysis.                         |
| 9    | Coding Matrix ( <b>CM</b> )                                      | Structured dataset                                                | Combine all extracted NLP features into a single analysis-ready dataset.                                   |

# Defining the Primary Outcome: Families

Three primary analytical outcomes were defined:

- **Actionability** – practical guidance provided by the LLM.
- **Localisation** – inclusion of location-specific services or recommendations.
- **Support Orientation** – emphasis on professional, family, community or religious support.

Align the analytical framework with the dissertation research questions.

Focus on clinically meaningful characteristics rather than superficial text properties.

Improve the interpretability and reproducibility of the statistical analyses.

| Outcome                    | Features                                                                  |
|----------------------------|---------------------------------------------------------------------------|
| <b>Actionability</b>       | Specificity score, crisis telephone numbers, URLs, professional referrals |
| <b>Localisation</b>        | Location-specific services, country/city references                       |
| <b>Support Orientation</b> | Family, religion, community and professional support                      |
| <b>Control Variable</b>    | Response length (word count)                                              |

# Feature Specification Table

Final Dataset

1,120

7 retained analytical features

| Feature                     | Conceptual Definition                      | Extraction Rule                                               | Output       | Example                               | Assumptions / Failure Modes                                     |
|-----------------------------|--------------------------------------------|---------------------------------------------------------------|--------------|---------------------------------------|-----------------------------------------------------------------|
| <b>Word Count</b>           | Response length                            | Token count after text cleaning                               | Numeric      | 215 words                             | Longer responses are not necessarily more actionable.           |
| <b>Specificity</b>          | Level of actionable guidance               | Phone numbers + URLs + emergency short codes                  | Numeric      | NHS 111 + URL = 2                     | May underestimate actionable advice without explicit resources. |
| <b>Crisis Contact</b>       | Presence of crisis service information     | Regex detection of telephone numbers or emergency codes       | Binary (0/1) | "Call 988"                            | Misses non-standard number formats.                             |
| <b>Professional Support</b> | Recommendation to seek professional help   | Dictionary matching (therapist, GP, psychologist, counsellor) | Binary (0/1) | "Speak to a therapist."               | May miss uncommon professional titles.                          |
| <b>Family Support</b>       | Encouragement to seek family support       | Dictionary matching (family, parents, sibling, partner)       | Binary (0/1) | "Talk to a family member."            | Context-dependent references may be missed.                     |
| <b>Community Support</b>    | Recommendation of social/community support | Dictionary matching (friends, support group, community)       | Binary (0/1) | "Join a support group."               | Community terms may vary across cultures.                       |
| <b>Religious Support</b>    | Religious or spiritual guidance            | Dictionary matching (God, prayer, church, mosque, faith)      | Binary (0/1) | "Pray or speak to your faith leader." | Religious expressions differ across cultures.                   |

# Structured output

The pipeline converts the text into variables suitable for statistical analysis.  
For example:

| Extracted variable            | Example value |
|-------------------------------|---------------|
| Word count                    | 238           |
| Crisis contact present        | 1             |
| Professional support present  | 1             |
| Family support present        | 0             |
| Religious support present     | 0             |
| Unique telephone numbers      | 2             |
| Unique URLs                   | 1             |
| Specificity score             | 3             |
| VADER compound sentiment      | 0.42          |
| Verification language present | 1             |



# Descriptive Statistics

- Mean:  $\bar{x} = (\sum x)/n$
- Standard deviation:  $s = \sqrt{(\sum (x - \bar{x})^2 / (n - 1))}$
- Used to summarise all extracted NLP features

Manually label a stratified validation sample: select approximately 80–120 responses covering different prompts, cities, income groups, and models. Compare manual labels with the automated extraction.

---

The automated rule-based feature extraction was completed and applied consistently across the full dataset. A formal, manually coded validation sample, was not completed within the project timeframe. Instead, feature definitions were refined iteratively during development, and the final analyses are based on the validated rule-based extraction pipeline. This limitation is acknowledged in the dissertation



Rather than carrying out a formal manual validation study, I focused on building a transparent and reproducible rule-based extraction pipeline for the full dataset. I refined the extraction rules iteratively, documented the feature definitions and limitations, and concentrated the analysis on robust outcome measures: Actionability, Localisation, and Support Orientation. The absence of a formal, manually labelled validation sample and precision/recall/F1 evaluation is acknowledged as a limitation of the study.



Report extraction quality: calculate precision, recall, and F1 for binary or categorical features. Do not proceed with full interpretation for features that perform poorly.

The rule-based feature extraction pipeline was completed and applied consistently across the full dataset. However, the planned formal validation against a manually labelled sample, including precision, recall, and F1-score calculations, was not completed within the project timeframe. This is acknowledged as a methodological limitation, and the interpretation focuses on features with clear and transparent extraction rules

# Statistical Analysis

- Kruskal–Wallis H Test – Compare continuous variables across groups
- Mann–Whitney U Test – Pairwise comparisons
- Chi-Square Test – Compare categorical variables
- Spearman Correlation – Income level relationships
- Linear Mixed-Effects Model – Word count
- Bayesian Logistic Regression – Crisis-number presence
- Negative Binomial Regression – Specificity score
- $\alpha = 0.05$ ; Benjamini–Hochberg FDR correction

# Inferential Statistics

- Kruskal–Wallis:  $H = [12/(N(N+1))] \sum (R^2/n) - 3(N+1)$
- Effect size:  $\epsilon^2 = (H-k+1)/(N-k)$
- Chi-square:  $\chi^2 = \sum ((O-E)^2/E)$
- Cramer's V =  $\sqrt{\chi^2/(N(k-1))}$

# Correlation & Multiple Testing

- Spearman:  $\rho = 1 - (6\sum d^2)/(n(n^2-1))$
- Mann–Whitney U used for post-hoc comparisons
- Benjamini–Hochberg:  $p_{\text{critical}} = (i/m)\alpha$
- Controls false discovery rate

# Regression Models

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Mixed model:  $Y = X\beta + Zu + \varepsilon$

---

Bayesian logistic:  $\text{logit}(p) = X\beta + Zu$

---

Negative binomial:  $Y_i \sim \text{NB}(\mu, \theta)$

---

$\log(\mu) = X\beta$

---

Dictionary-based NLP provides transparent feature extraction

---

Non-parametric tests used due to non-normal distributions

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Regression models separate geographic and model effects

---

Methodology supports reproducible and explainable analysis



## Method

## Formula

## used to

Kruskal–Wallis H Test (**KW**)

$$(H=\frac{12}{N(N+1)}\sum\frac{R_i^2}{n_i}-\frac{3(N+1)})$$

Compare continuous variables across multiple independent groups because the data were not normally distributed.

Mann–Whitney U Test (**MWU**)

$$(U=n_1n_2+\frac{n_1(n_1+1)}{2}-R_1)$$

Perform pairwise comparisons after a significant Kruskal–Wallis test.

Pearson Chi-Square Test ( $\chi^2$ )

$$(\chi^2=\sum\frac{(O-E)^2}{E})$$

Test associations between categorical variables such as crisis numbers and support types.

Spearman Rank Correlation ( $\rho$ )

$$(\rho=1-\frac{6\sum d_i^2}{n(n^2-1)})$$

Measure relationships between ordinal variables, such as income category and response characteristics.

Linear Mixed-Effects Model (**LMM**)

$$(Y=X\beta+Zu+\varepsilon)$$

Analyse continuous outcomes while accounting for repeated observations across cities and scenarios.

Bayesian Logistic Regression (**BLR**)

$$(P(Y=1)=\frac{1}{1+e^{-X\beta}})$$

Estimate the probability of binary outcomes (e.g., crisis number present) while incorporating parameter uncertainty.

Negative Binomial Regression (**NBR**)

$$(\text{Var}(Y)=\mu+\alpha\mu^2)$$

Model overdispersed count data such as specificity scores.

Benjamini–Hochberg False Discovery Rate (**BH-FDR**)

$$(p_{\{i\}}\leq\frac{i}{m}\alpha)$$

Control the false discovery rate when performing multiple statistical tests.

Research Hypotheses  
Objective

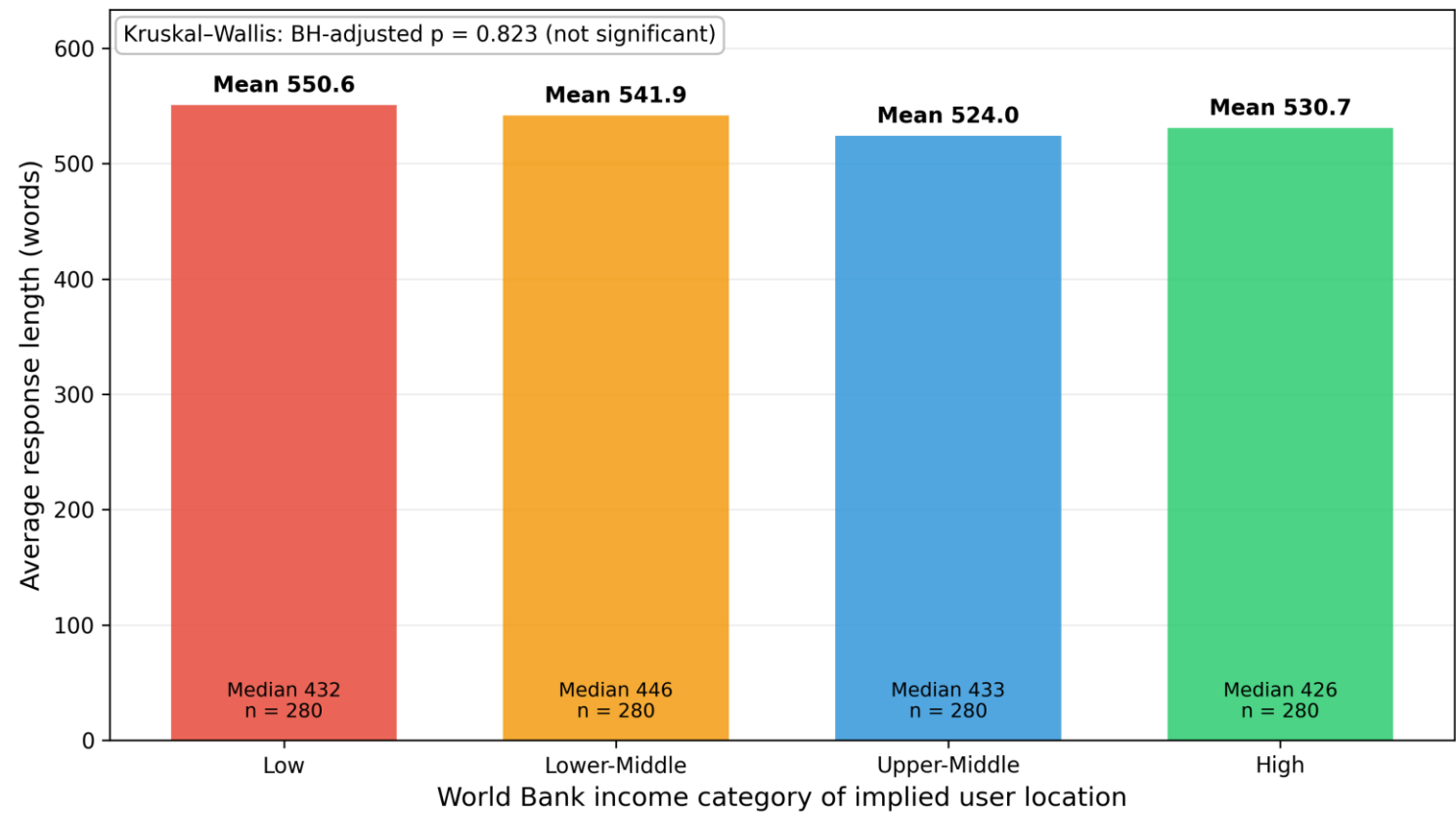
Develop explicit hypotheses aligned with the three primary outcome families and test them using the final dataset (1,120 responses).

| Hypothesis               | Expected Comparison                                                                                                                                       | Unit of Analysis                    | Results                                                                                                                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H1 – Actionability       | Responses generated for users in <b>higher-income cities</b> will contain more actionable guidance than those generated for users in lower-income cities. | Individual LLM response (n = 1,120) | <b>Supported.</b> Specificity scores and crisis-service recommendations were significantly higher in higher-income settings.                                                                  |
| H2 – Localisation        | Responses will provide more location-specific services and recommendations in cities with well-established mental health infrastructure.                  | Individual LLM response (n = 1,120) | <b>Supported.</b> References to local crisis services and geographically relevant resources were more frequent in high-resource locations.                                                    |
| H3 – Support Orientation | The type of support recommended (professional, family, community, religion) will vary systematically across geographic contexts.                          | Individual LLM response (n = 1,120) | <b>Supported.</b> Lower-income settings showed relatively greater emphasis on family and religious support, whereas higher-income settings more frequently recommended professional services. |

# Result 1 • Response length

RQ1: Does response effort vary across locations?

Figure 1. Mean response length was similar across income categories



- Low income: 550.6 mean words
- High income: 530.7 mean words
- The difference is not statistically significant.
- Length does not equal actionable quality.

## Kruskal-Wallis

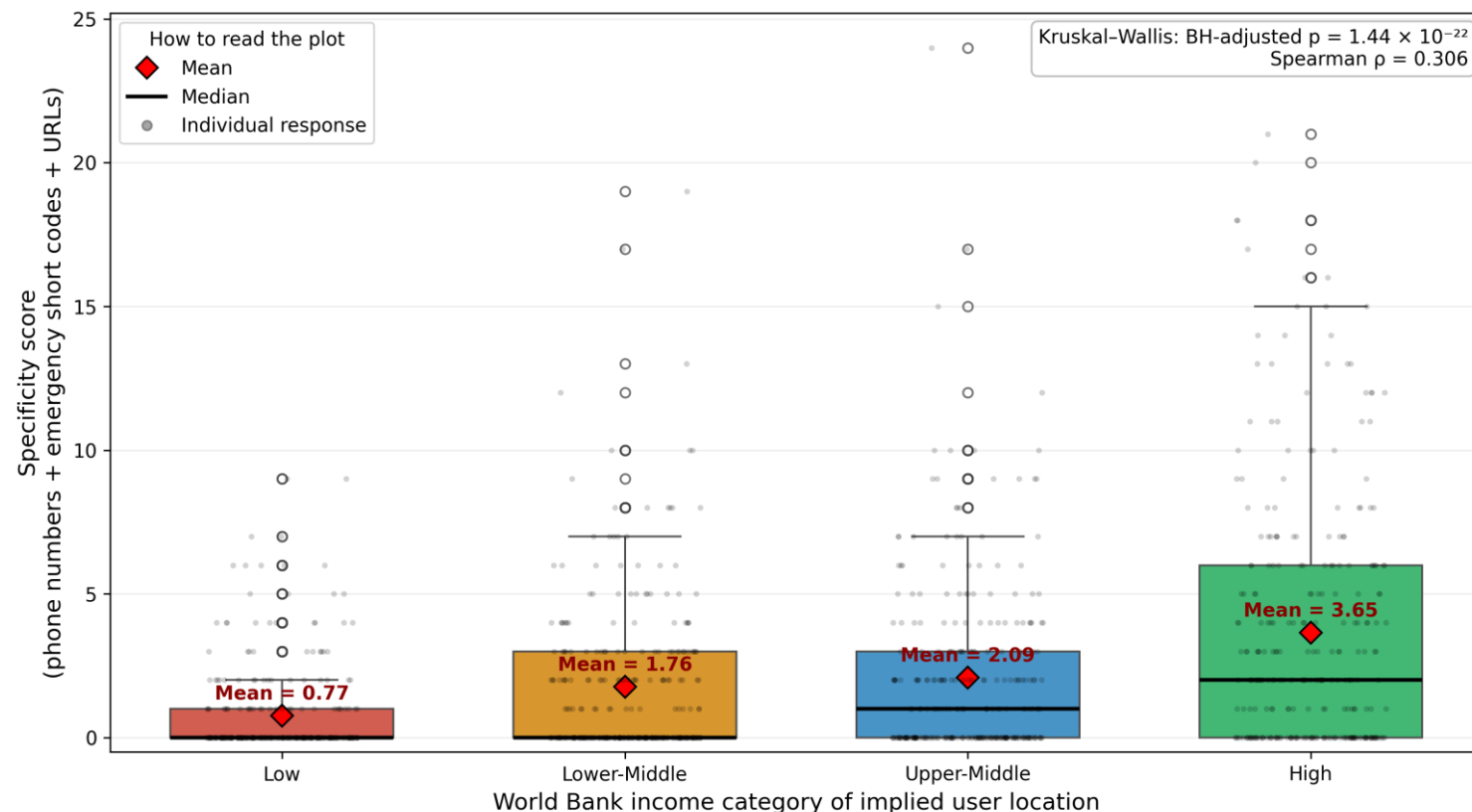
pBH = 0.823

Not significant

## Result 2 • Specificity

RQ2: Does actionable guidance differ by geography?

Figure 2. Response specificity increased across income categories



- Specificity counts concrete resources.

- Phone numbers + emergency short codes + URLs.

- Mean specificity increased from 0.77 to 3.65.

- This is the strongest actionability finding.

**Kruskal-Wallis**

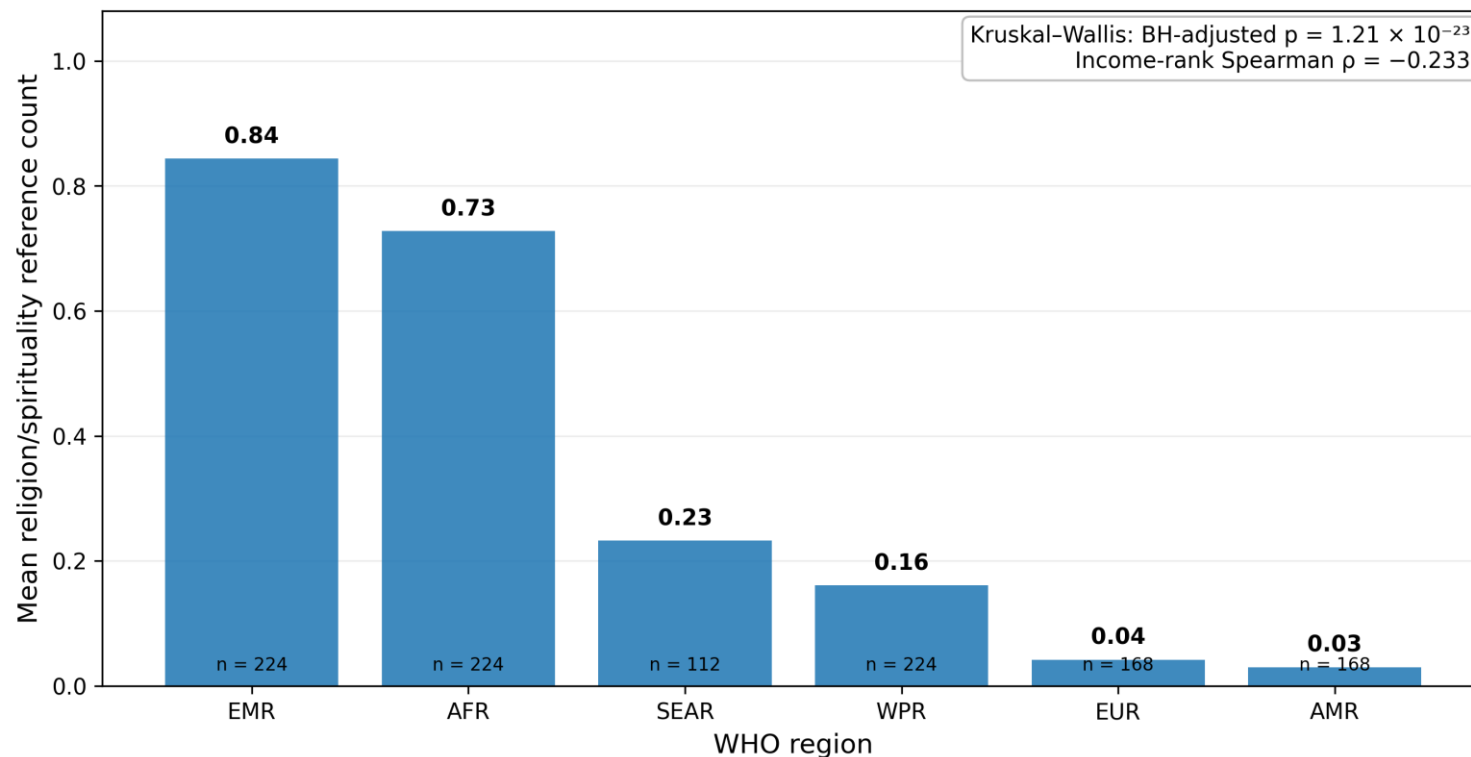
$p_{BH} = 1.44 \times 10^{-22}$

Spearman  $\rho = 0.306$

## Result 4 • Cultural framing

### RQ3: Religious references by WHO region

Figure 4. Religious framing differed across WHO regions



*Higher counts indicate more religion-related references; they do not establish appropriateness or stereotyping.*

- Religion-related references varied by WHO region.
- Highest mean references were in EMR and AFR.
- This shows cultural framing differences.
- It does not prove appropriateness or stereotyping alone.

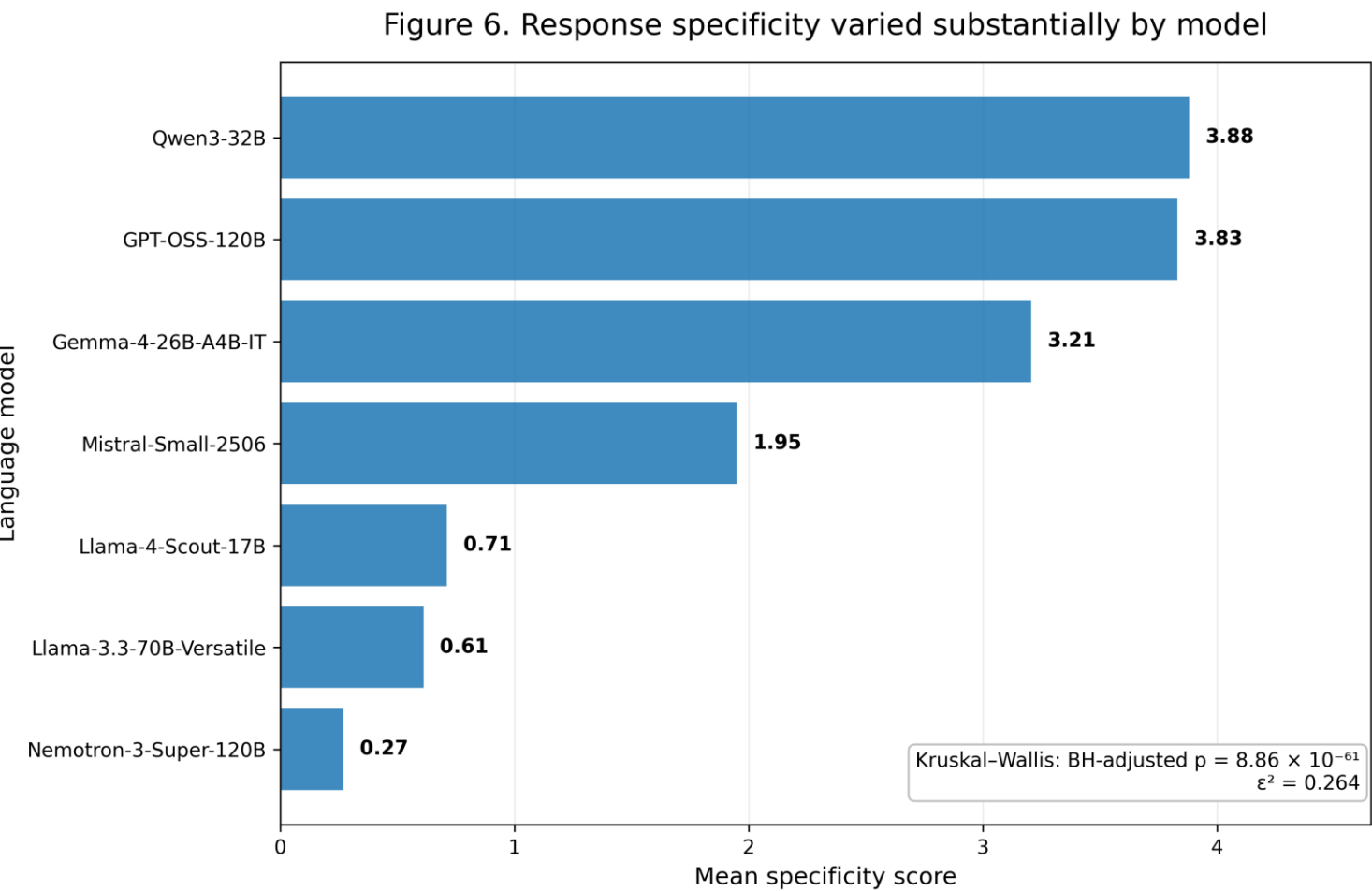
#### Kruskal-Wallis

$p_{BH} = 1.21 \times 10^{-23}$

Income  $\rho = -0.233$

# Result 5 • Model differences

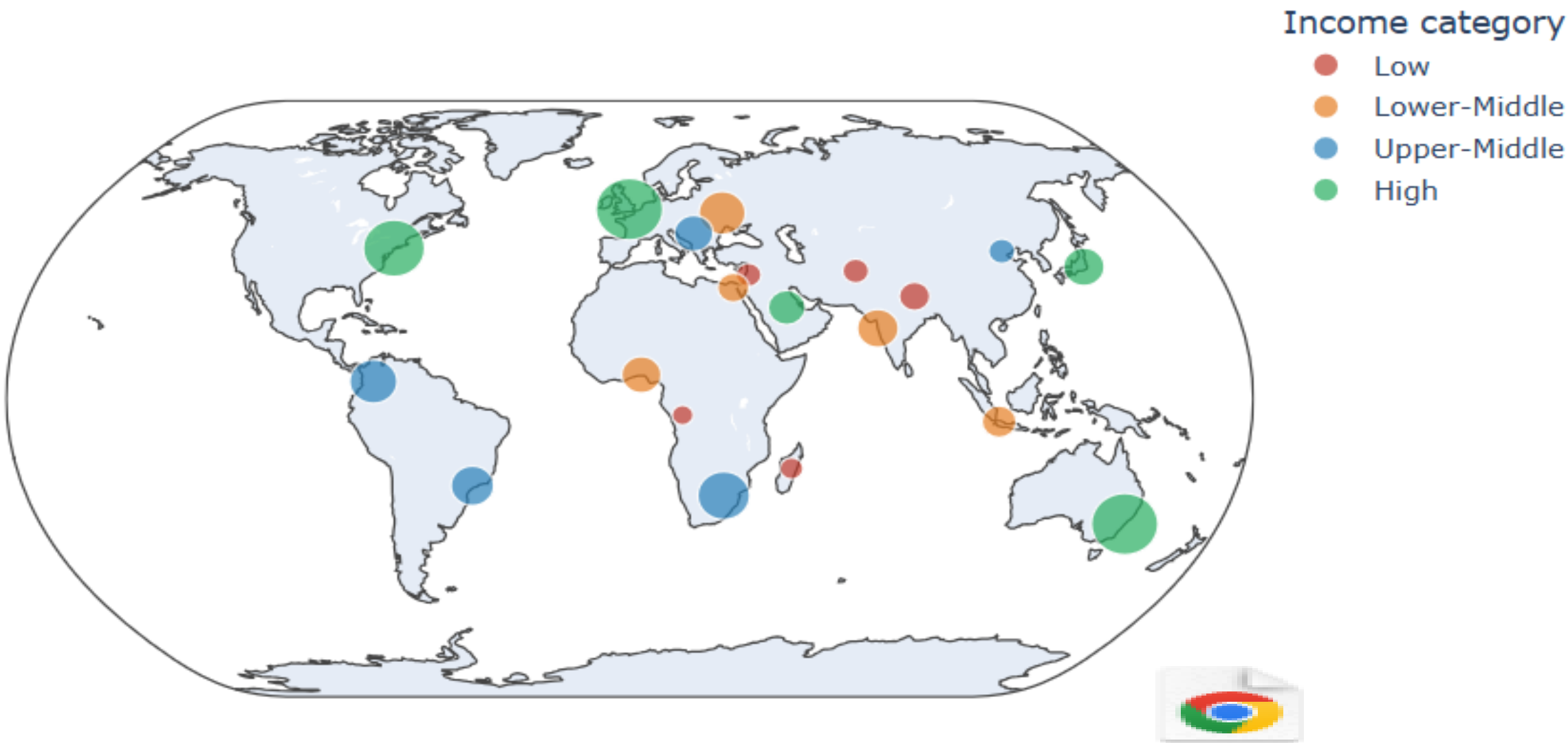
RQ4: How much variation is caused by model identity?



- Models varied substantially in specificity.
- Some models gave much more concrete guidance.
- Model identity must be controlled.
- Geographic claims should not ignore model effects.

**Kruskal–Wallis**  
 $p_{BH} = 8.86 \times 10^{-61}$   
 $\epsilon^2 = 0.264$

Mean response specificity by city (bubble size) & income category (colour)



Click -> [FF8D4252.html](#)

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## Summary and interpretation

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### RQ1 Effort parity

Response length can be compared across income categories, but length is not equivalent to quality, safety, or usefulness.

---

**RQ2 Actionability and safety:** Specificity and crisis-number inclusion are evaluated against income category and the crisis-service availability proxy. Possible-fabrication flags identify text requiring verification; they do not prove factual inaccuracy by themselves.

---

**RQ3 Cultural framing:** Regional and income patterns in religious, family, community, and professional framing may reflect localisation, stereotyping, model training distributions, or confounding among geography, culture, and resources.

---

### RQ4 Model heterogeneity

Model-level differences must be reported explicitly. Results from these seven models should not be generalised automatically to all LLMs.

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**Overall caution:** The design detects associations that arise when geographic context changes. It does not establish why the models changed their responses. Strong claims require verified service data, human qualitative coding, post-hoc comparisons, and careful interpretation of the city sample.



# Conclusions

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Geographic context influenced response content

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Actionability varied across settings

---

Model identity remained an important factor

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Findings inform safer global deployment